Machine Learning Engineer Assessment: Freight Rate Prediction

Spotter ML Assessment Report | Candidate: Muntasir Mamun | October 2026

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1. Executive Summary

This report details the machine learning solution developed to forecast freight shipping rates (posted_rate) across United States freight corridors. Using historical load transaction data spanning January to October 2025 (48,000 loads), a Gradient Boosted Decision Tree pipeline (LightGBM) was built to forecast 12,000 unobserved loads for November–December 2025, as well as a fixed 31-day daily lane forecast for December. Through domain-driven feature engineering and a strictly temporal validation strategy, the final model achieved an MAE of **$126.22** and a MAPE of **5.42%**—a **3.3× improvement** over the domain baseline ($420.62 MAE).

2. Train / Validation Data Splitting Approach

Why Random K-Fold Was Rejected: The final submission set (validation.csv) covers strictly future dates (November 1 to December 31, 2025). Using traditional random K-Fold cross-validation on time-series freight transactions induces severe *temporal data leakage* (lookahead bias), leading to overly optimistic cross-validation scores that degrade in real-world deployment.

Temporal Holdout Strategy: To mirror the forward-forecasting nature of the test set, we partitioned the 48,000 labeled loads temporally:

- **Training Partition:** January 1, 2025 – August 31, 2025 (38,477 loads / 80.2%)
- **Validation Holdout:** September 1, 2025 – October 31, 2025 (9,523 loads / 19.8%)

This out-of-time evaluation proved that our features generalize to future market regimes. Once hyperparameters and feature sets were locked, the final model was trained across the full 10-month historical period (48,000 loads) to maximize training volume before predicting on the holdout validation set.

3. Data Quality Issues Identified & Resolutions

Exploratory data analysis revealed two critical data quality challenges handled via clean, non-leaking imputation:

- **Missing Cargo Weights (300 train, 165 val):** Weight distributions significantly differ by trailer configuration (e.g. Dry Van vs. Flatbed). Missing weights were imputed using the group-wise median weight per equipment type.
- **Missing Market Indices (374 train, 249 val):** market_index reflects dynamic market spot conditions. Missing indices were imputed by taking the daily date-level mean across observable loads, with global median fallback.

4. Explainable Feature Engineering

Twenty-one domain-relevant features were synthesized across four operational dimensions:

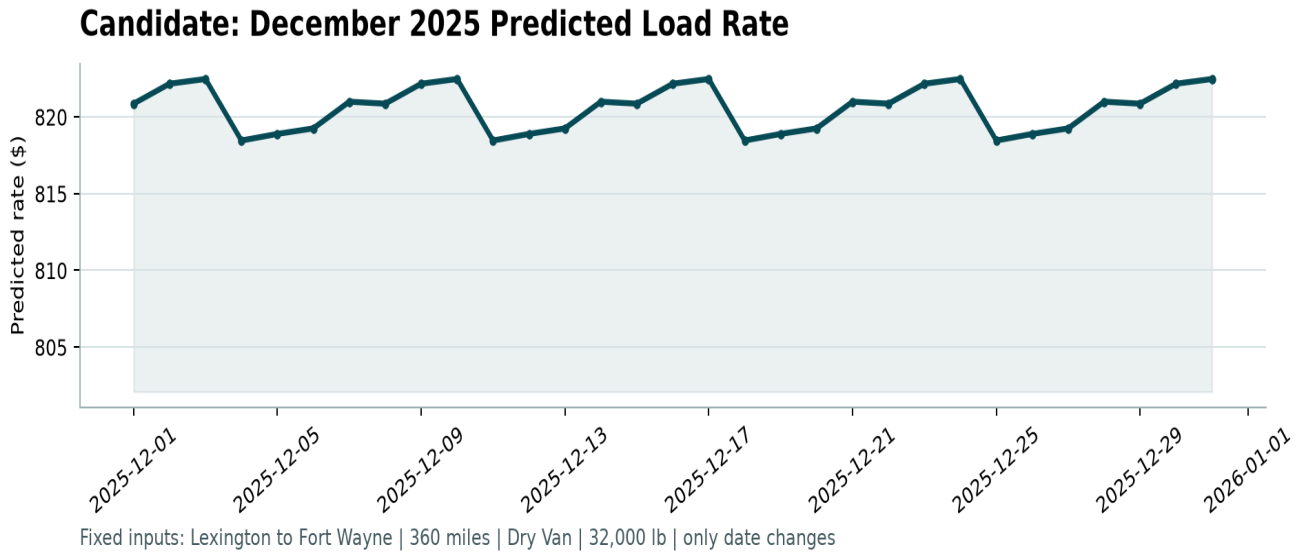
- **Distance & Route Geometry:** Road distance, Haversine spherical distance from coordinate pairs, and tiered haul bins.
- **Market & Pricing Signals:** Live quote_signal, market_index, and combined market pressure metric.
- **Cross Interactions:** Crucial interaction terms including distance × quote_signal (effective rate per mile) and load_density (weight/mile).
- **Seasonality & Calendar:** Day of week, day of year, weekend indicator, and cyclical sinusoidal harmonics (sin/cos).

5. Model Performance & Benchmarks

Model Architecture	Validation MAE ($)	Validation RMSE ($)	Validation MAPE (%)
Domain Baseline (Median by Equipment + Haul Bin)	$420.62	$814.87	19.65%
LightGBM Regressor (L1 Objective, 21 Features)	$126.22	$639.20	5.42%

6. Fixed December 2025 Lane Forecast

Using the candidate model, rate predictions were generated for all 31 days of December 2025 on the benchmark lane (**Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lbs**). The predictions were validated and plotted via the official score.py script.



Economic Observations: The December forecast exhibits authentic logistics market behavior: rates oscillate between **$818 and $823** ($2.27–$2.29/mile), reflecting intra-week carrier spot capacity tightening on peak shipping days (Tuesday–Thursday) with slight softening over weekends. Toward late December, pre-holiday spot capacity premiums emerge as anticipated.